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Engineering Design 101: The Paper Tower Challenge — A Full Design Session

Engineering Design · Engineering / CTE

COURSE-OBJ.ENGR101.2

Apply the engineering design process to define a problem, plan a fair test, and justify a design iteration using test evidence rather than guesswork. A complete 90-minute session packet with warm-up, guided practice, partner/lab work, independent practice, extension work, assessment, and a full answer key.

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Plan the session

Course and objective

Engineering Design · Engineering / CTE; COURSE-OBJ.ENGR101.2 — Define a design problem, distinguish criteria from constraints, generate and compare concepts using a weighted decision matrix, plan a fair test, and justify an evidence-based iteration.

What is included

Cover; instructor guide; key-term and concept preview; original reading or worked examples; warm-up (2 items); guided practice with organizer (4 items); partner/lab discussion (3 items); independent practice (6 items); small-group extension work (4 items); assessment; complete answer key; access and sources page.

Suggested pacing (90-minute session)

Warm-up 8 min · read both passages + vocabulary 15 min · guided criteria/constraints organizer 18 min · partner evidence discussion 18 min · independent practice 22 min · extension work 5 min · assessment 4 min.

Preparation

Print student pages separately from the answer key, or distribute through a secure LMS. Review the reading/examples and organizer before class. No paid textbook or app is required.

Student directions

Read or work through the material carefully. Support every answer with evidence, work shown, or clear reasoning.

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Before you begin

Key terms

criterion — a desired feature or measure of success

constraint — a hard limit the design must obey

prototype — a testable representation of a design idea

fair test — holding every variable constant except the one studied

iteration — a targeted, evidence-justified design change

decision matrix — a weighted scoring table comparing design concepts

Concept focus

A criterion is a desired feature or measure of success ("must support 500 g"); a constraint is a limit the design must obey ("no more than 10 sheets of paper"). Confusing the two leads teams to "solve" the wrong problem. A fair test holds every variable constant except the one being studied, so a failure can be traced to a specific cause. A weighted decision matrix scores each concept against criteria that are not all equally important, multiplying each score by a weight before totaling. Good engineers do not treat a failed prototype as a dead end — they record exactly where and how it failed, then use that evidence to justify one specific, targeted change for the next iteration.

Model 1 — Sorting criteria from constraints

A design brief says: "Build a free-standing tower using no more than 10 sheets of paper and one 12-inch strip of tape, that supports at least 500 g at 30 cm of height, built within 25 minutes." The paper/tape limits and the time limit are constraints (hard limits); "supports at least 500 g" and "at least 30 cm tall" are criteria (measures of success).

Model 2 — Reading a weighted decision matrix

Three concepts are scored 1-5 on two criteria: Strength (weight x3) and Build speed (weight x1). Concept A scores 4 strength, 5 speed: (4x3)+(5x1)=17. Concept B scores 5 strength, 2 speed: (5x3)+(2x1)=17. Even with a tie, the matrix makes the tradeoff visible instead of hidden.

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Warm-up

Answer using what you already know or recall from the previous session.

1. In the design brief, list two constraints and explain why they are constraints and not criteria.

2. List two criteria from the design brief and explain why they are criteria and not constraints.

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The Design Brief: A Paper Tower Challenge

The assignment is simple to state and hard to solve well: using no more than 10 sheets of printer paper and one 12-inch strip of tape, build a free-standing tower that supports a textbook (about 500 grams) balanced flat on top, reaches at least 30 centimeters tall, and fits within a 20-by-20-centimeter footprint on the table. Teams have 25 minutes to design, build, and test, and must record every test result, not just the final one.

Before any paper is folded, each team is required to sort the brief into criteria and constraints on a shared worksheet. The paper limit, the tape limit, the footprint, and the time limit are all constraints: hard limits the design must obey no matter what. The height and the load are criteria: measures the team is trying to hit or beat, and a design that supports 700 grams at 35 centimeters has exceeded the criteria, not just met them. A team that skips this sorting step often wastes early minutes arguing about a limit that was never actually fixed, or, worse, discovers with five minutes left that their beautiful tower is two centimeters short of the height criterion.

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Two Teams, Two Approaches

Team Turing rolls all 10 sheets into a single wide column, tapes the seam, and stands it up in under four minutes — by far the fastest build in the room. It easily clears the 30-centimeter height criterion. But when they lower the textbook onto it, the column buckles sideways within two seconds and the book slides to the floor. They try again with the same column shape, taping a second seam for reinforcement; it buckles again, in the same spot, just slightly slower. Their build log shows two failed trials with the same failure mode: no lateral bracing.

Team Lovelace spends longer planning: they fold four sheets into triangular corner posts, using the geometric fact that a triangle resists changing shape under load far better than a plain cylinder or rectangle does, and use the remaining sheets to form horizontal cross-braces at two heights, widening the base for stability. Their first test holds 500 grams but tips slightly when the book is placed off-center. Reading that result, they widen the base by two centimeters and add one more low cross-brace — a small, targeted change justified directly by where and how the first prototype failed, not a full redesign. The second test holds the book centered and off-center alike, and the structure is measured at 31 centimeters, one centimeter above the height criterion.

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Guided practice · Reasoning organizer

Work through this organizer together before independent practice.

Given / Evidence	Method / Reasoning	Conclusion
State the given information, quote, or data.	Explain the method or reasoning in your own words.	State the answer or claim it supports.

1. Why does the lesson say a team that skips sorting criteria from constraints "often wastes early minutes"?

2. Using Team Turing's two trials, identify the controlled variable(s) and the one thing that changed between trials.

3. What is a "fair test," and how does Team Turing's second trial fail to isolate a new variable?

4. Explain, using Team Lovelace's reasoning, why triangles are used in the design instead of a plain column.

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Partner / lab discussion

Discuss each item with a partner or lab group before writing a joint response.

1. With a partner, discuss: was Team Turing's first build a failure? Use the lesson's framing of prototypes to justify your answer.

2. With a partner, identify exactly what evidence justified Team Lovelace's second-iteration change (widening the base, adding a brace). Was the change targeted or a full redesign?

3. Discuss: which team's process better matches the engineering design process described in the lesson, and why?

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Independent practice

Work independently. Show your work or cite evidence in every answer.

1. Create a two-criterion, two-concept weighted decision matrix using Strength (weight x3) and Cost (weight x2), with your own 1-5 scores, and calculate both totals.

2. Explain in your own words why a weighted decision matrix is more useful than simply picking a "favorite" concept.

3. Distinguish accuracy from precision, then explain which one repeated identical test trials mainly demonstrate.

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Independent practice

Work independently. Show your work or cite evidence in every answer.

4. Propose one specific safety rule for this challenge and explain the specific risk it addresses.

5. Write a testable claim about paper column strength that a team could investigate with a fair test.

6. Using Team Turing's data, write one sentence of evidence-based reasoning that could justify their next iteration (which the passage does not describe).

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Small-group extension work

Use for review or additional application practice.

1. Reflect on tradeoffs: if Team Lovelace had used all 10 sheets for cross-bracing and none for height, what criterion would likely suffer, and why?

2. Explain why recording every test result, not just the final one, matters to the engineering design process described in this lesson.

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Small-group extension work

Use for learners ready for additional challenge or synthesis.

1. Design, in words, one original modification to the paper tower challenge (a new criterion or constraint) and explain what new tradeoff it would create.

2. Write a short paragraph (4-5 sentences) comparing Team Turing's and Team Lovelace's approaches, using at least three vocabulary terms from today's lesson.

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Assessment

Complete independently to show what you learned today.

1. Self-assessment: name one part of the engineering design process you feel confident applying, and one you want more practice with.

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Assessment · self-assessment

Rate today’s work:

4 — I can teach this to someone else.

3 — I can do this independently.

2 — I can do this with help.

1 — I need to revisit this before the next session.

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Answer key

1. The 10-sheet/one-tape-strip material limit and the 25-minute time limit are constraints because they are hard limits the design must obey; they are not things to maximize or optimize, unlike criteria.

2. Supporting at least 500 g and reaching at least 30 cm tall are criteria because they are measures of success a team is trying to meet or exceed, not fixed limits that cannot be crossed.

3. Without sorting them first, teams may argue about a limit that was never actually fixed, or discover too late that they never confirmed what "success" required, wasting time that could have gone to building and testing.

4. Controlled: the overall column shape/approach and the load tested. Changed: a second taped seam was added as reinforcement between trial 1 and trial 2 — but the underlying shape (no lateral bracing) stayed the same, which is why the same failure mode repeated.

5. A fair test holds every variable constant except the one being studied. Team Turing's second trial changed a seam but kept the same unbraced column shape, so it did not test a genuinely different design idea — it retested a very similar one and got a very similar failure.

Accept equivalent, well-supported alternative answers throughout.

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Answer key

6. A triangle resists changing shape under load better than a plain cylinder or rectangle; Team Lovelace uses this geometric property to brace their corner posts and cross-braces, giving the structure lateral stability that Team Turing's column lacked.

7. By the lesson's framing, a failed prototype is not a dead end but evidence — Team Turing's first build revealed a specific failure mode (buckling from no lateral bracing), which is useful information even though the prototype itself did not meet the load criterion.

8. The evidence was that the first prototype held the load but tipped when off-center; the change (widening the base, adding one low brace) directly targets that specific instability rather than redesigning the whole structure — a targeted, evidence-based iteration.

9. Team Lovelace better matches it: they distinguished criteria from constraints in their planning, used a structural principle (triangulation) deliberately, tested, and made one targeted, evidence-justified change rather than repeating the same untested assumption.

10. Answers vary; must show two concepts each scored 1-5 on both criteria, multiplied by the correct weights (x3 and x2), with correctly calculated totals shown as work.

Accept equivalent, well-supported alternative answers throughout.

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Answer key

11. A weighted matrix makes tradeoffs explicit and numeric rather than hidden in a gut feeling, and it lets a team weigh criteria that matter more (like strength) more heavily than criteria that matter less (like build speed), producing a more defensible decision.

12. Accuracy is closeness to the true value; precision is repeatability of results. Running repeated identical trials mainly demonstrates precision (consistency), though it can also help reveal whether results are accurate if compared to a known standard.

13. Answers vary; must name a specific rule (e.g., "keep test loads under 1 kg and test at table height, not overhead") tied to a specific risk (e.g., a falling weight or a collapsing structure near hands/eyes).

14. Answers vary; must state a specific, testable comparison, e.g., "A triangulated paper column will support more weight than a plain rolled column of the same paper and height," with an implied fair-test setup (same paper, same height, only the shape changes).

15. Answers vary; must cite the specific failure evidence (buckling from lack of lateral bracing) and propose a targeted change addressing that exact cause, e.g., "Because both trials buckled sideways with no bracing, add diagonal cross-braces at the midpoint."

Accept equivalent, well-supported alternative answers throughout.

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Answer key

16. The height criterion (at least 30 cm) would likely suffer, because using all the material for bracing near the base would leave little paper to build the structure taller — strengthening one criterion (stability) at the direct expense of another (height).

17. Every result is evidence; recording failed trials (like Team Turing's two identical buckling failures) is what makes it possible to justify a specific next change rather than guessing, and shows whether a change actually fixed the identified problem.

18. Answers vary; must propose a specific new criterion or constraint (e.g., "must also survive a small fan blowing on it for 10 seconds") and explain a real resulting tradeoff (e.g., "a wider, heavier base would help but could reduce height achievable with the same material").

19. Answers vary; must be a coherent comparison using at least three terms (e.g., criterion, constraint, fair test, iteration, prototype) accurately applied to the two teams' described approaches.

20. Answers vary; must name a specific part of the process described in the lesson (e.g., sorting criteria/constraints, building a decision matrix, planning a fair test, justifying an iteration) for each part.

Accept equivalent, well-supported alternative answers throughout.

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Access, sources, and license

Extension

Ask for a second method, a counterexample, or a short student-led extension of the concept, with a brief explanation of the choice.

Accommodated seating/504

Follow the individual accommodation plan: extended time, breaks, enlarged print, reduced copying, or an alternate response method as documented.

Additional support

Pre-teach key vocabulary or notation before class. Model the first guided item aloud. Accept oral or dictated responses when appropriate.

Multilingual learners

Preview key terms with examples. Allow brief first-language discussion before independent writing. Provide sentence or solution starters.

Course-level alignment · accessed September 18, 2026

COURSE-OBJ.ENGR101.2 — Define a design problem, distinguish criteria from constraints, generate and compare concepts using a weighted decision matrix, plan a fair test, and justify an evidence-based iteration. The objective is original wording describing a focused part of the course-level goal.

Course-level learning objective written for this packet; no accreditation, transfer credit, or universal course-equivalency is claimed.

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